

Tes-01 · Dual-CAN Gateway

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Technical datasheet

STM32F405 dual-CAN bus board with Wi-Fi (ESP32-S3) · automotive 12–16 V input Rev v3 · 2-layer · 62 × 48 mm

Overview

The **Tes-01** is a compact, ready-to-use **two-channel CAN-bus interface board** built around an **STM32F405RGT6** (Cortex-M4, 168 MHz) with two **SN65HVD230** high-speed CAN transceivers and an on-board **Seeed XIAO ESP32-S3** Wi-Fi/Bluetooth module. It runs directly from a **12–16 V automotive supply** (reverse-polarity and transient protected) and exposes both CAN channels on screw-terminal blocks. The ESP32-S3 turns it into a **Wi-Fi gateway**: a built-in web dashboard lets you monitor and configure the board from a phone or PC, and the STM32 firmware can be **updated over Wi-Fi** — no external programmer required.

It is an open, fully user-programmable development board. You write and load your **own firmware** into both the STM32 and the ESP32-S3 with standard free toolchains — the board ships with ready-to-run example firmware only as a starting point, not as a locked product.

Fully user-programmable (open)

Link up to 1 Mbit/s

2× independent CAN @ 500 kbit/s

STM32F405 · M4 168 MHz

Wi-Fi + BLE (ESP32-S3)

12–16 V automotive input

Reverse-polarity + TVS

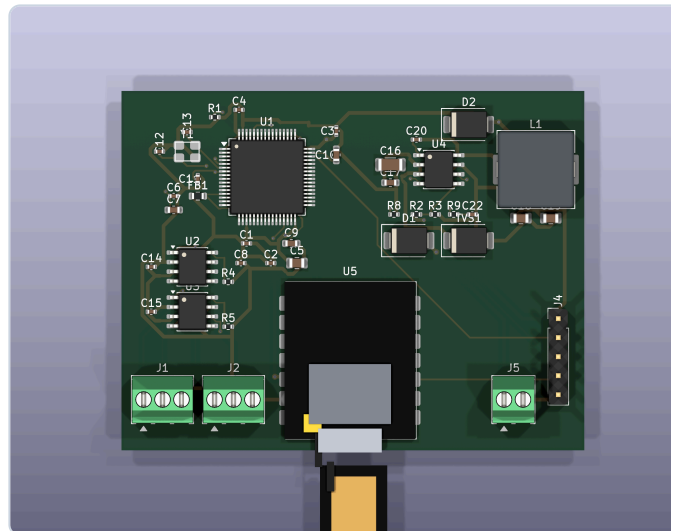
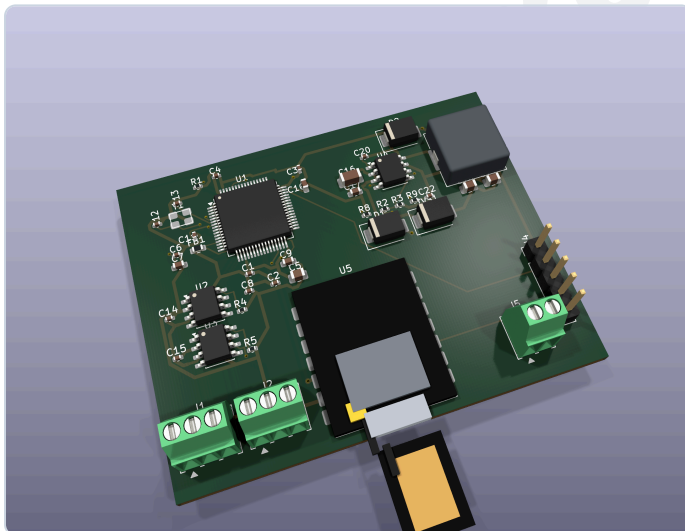
Web dashboard

OTA firmware update

RDP1 code protection over Wi-Fi

SWD header

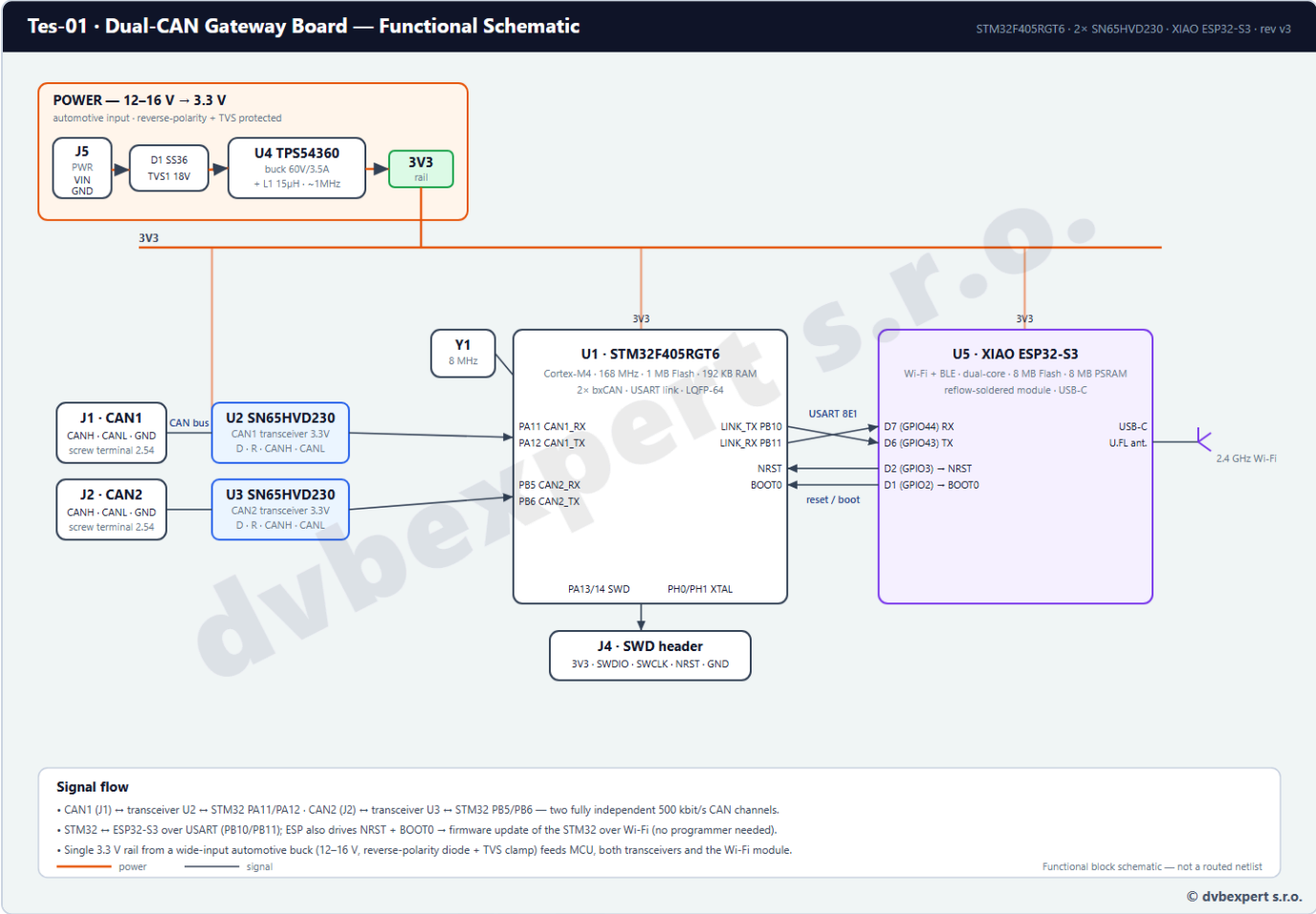
62 × 48 mm



Typical applications

- CAN-bus development, prototyping and bench testing
- Two-bus **gateway / bridge** between independent CAN segments
- CAN traffic monitoring & data logging over Wi-Fi
- Vehicle / machine telemetry to a phone, PC or cloud
- Retrofit & IoT add-ons that read or relay CAN data
- Automation, robotics and education / STEM projects

Block schematic

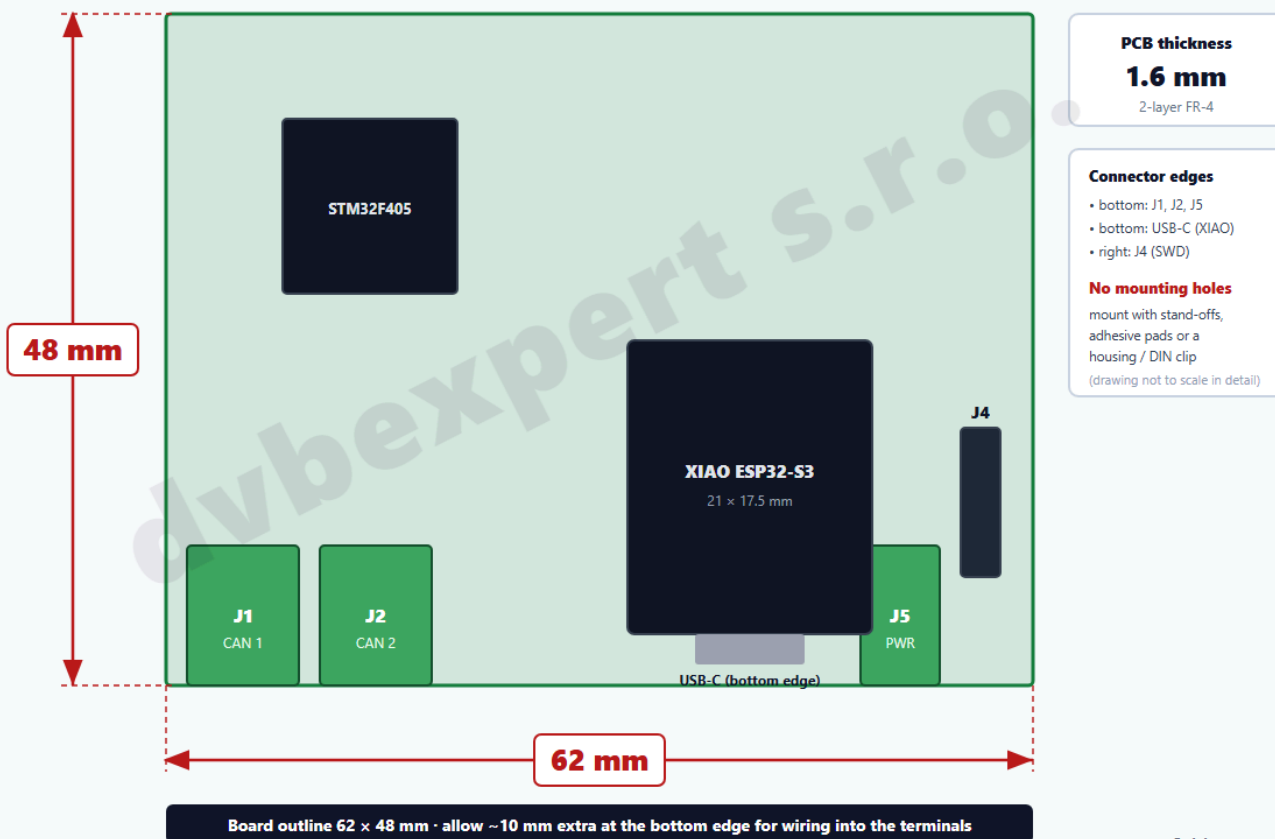


Technical specifications

Microcontroller	STMicroelectronics STM32F405RGT6 — ARM Cortex-M4F, 168 MHz, 1 MB Flash, 192 KB SRAM, LQFP-64
CAN channels	2× independent bxCAN, ISO 11898-2 high-speed, default 500 kbit/s (firmware-configurable)
CAN transceivers	2× Texas Instruments SN65HVD230 (3.3 V, up to 1 Mbit/s)
Isolation	None — non-isolated, common ground (designed to tap existing buses)
Wireless module	Seeed XIAO ESP32-S3 — Wi-Fi 802.11 b/g/n 2.4 GHz + Bluetooth LE, dual-core, 8 MB Flash, 8 MB PSRAM, U.FL antenna
MCU ↔ Wi-Fi link	USART 8E1, 115200 default — up to 1 Mbit/s (interrupt-driven RX, verified on hardware); ESP32-S3 also controls STM32 <code>NRST</code> + <code>BOOT0</code> for wireless firmware update
Power input	12–16 V DC (automotive), reverse-polarity Schottky + SMBJ18A TVS transient/surge clamp
Regulator	TI TPS54360 wide-input buck (60 V / 3.5 A) → single 3.3 V rail (MCU + transceivers + Wi-Fi)
Clock	8 MHz crystal (SMD 3225) → 168 MHz PLL
Debug	5-pin SWD header (3V3 · SWDIO · SWCLK · NRST · GND)
USB	USB-C on the ESP32-S3 module (power + serial + first-time flashing)
Board	2-layer FR-4, 62 × 48 mm , solid ground plane, SMD assembled + XIAO reflow-soldered
Connectors	3× 2.54 mm terminal blocks (2× CAN, 1× power) + 1× 5-pin SWD header

Mechanical — board dimensions

PCB outline	62 × 48 mm (2.44 × 1.89 in)
PCB thickness	1.6 mm , 2-layer FR-4, 1 oz copper
Connector edges	Bottom edge: J1 (CAN1), J2 (CAN2), J5 (power) + USB-C of the XIAO module · Right edge: J4 (SWD)
Mounting holes	None — mount with stand-offs, adhesive pads or inside a housing
Clearance	allow ≈ 10 mm beyond the bottom edge for wires entering the screw terminals



Connector pinout

Connector	Pin 1	Pin 2	Pin 3
J1 — CAN1	CAN1_H	CAN1_L	GND
J2 — CAN2	CAN2_H	CAN2_L	GND
J5 — Power	VIN (12–16 V)	GND	—
J4 — SWD	5-pin ST-Link programming header — see the table below		

⚠ IMPORTANT — no on-board CAN termination. The board ships **without** 120 Ω bus termination by design, because it is meant to tap an **existing, already-terminated** bus (a vehicle bus is typically ~60 Ω).

- **In a vehicle / on a live bus:** do **NOT** add termination — the bus already has it.
- **On the bench (stand-alone test):** you **MUST** add **120 Ω between CAN H and CAN L**, otherwise the bus never returns to the recessive state and the controller reports stuff / bit errors and goes bus-off.

bottom edge of the board (where the terminal blocks are)

**Wiring summary****J1 — CAN 1**

pin 1 = CAN H (HIGH) · pin 2 = CAN L (LOW) · pin 3 = GND

J2 — CAN 2

pin 1 = CAN H (HIGH) · pin 2 = CAN L (LOW) · pin 3 = GND

J5 — POWER

pin 1 = +12~16 V DC · pin 2 = GND (–) — reverse-polarity protected

Notes

- Pin order shown as seen from the TOP (component) side, left → right.
- NO on-board 120 Ω termination — for a BENCH TEST add 120 Ω between CAN H and CAN L.
- CAN H / CAN L must go to the matching H / L of the bus.
- GND of J1/J2 is the same board ground as J5 pin 2.
- Powering: connect +12~16 V to J5 pin 1, vehicle/supply ground to J5 pin 2.

Tes-01 · connector reference · matches board silk J1/J2/J5/J4

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Connector wiring reference — view from the top (component) side, left → right.

Programming & development (program it yourself)

Both processors are programmed with **standard, free toolchains** — this is an open board for your own applications:

Chip	Program it in	How to load
STM32F405	C / C++ (CMSIS, HAL/LL), Rust, or Arduino_Core_STM32	over Wi-Fi (ROM bootloader, AN3155) · or SWD / ST-Link on header J4
ESP32-S3	Arduino-ESP32 or ESP-IDF (C/C++)	USB-C on the module · or Wi-Fi OTA

The pre-loaded gateway firmware (web dashboard, OTA, CAN self-test) is an **example / starting point** — replace it with your own at any time. The two CAN channels, the STM32↔ESP link and the reset/boot control are all yours to use.

Performance / throughput

- CAN runs at **500 kbit/s** by default (firmware-configurable up to 1 Mbit/s). At 500 kbit/s the bus carries up to **~4500 full frames/s**; e.g. **2000 msg/s ≈ 44 % bus load** — well within range.
- The **STM32F405 @ 168 MHz** with two hardware bxCAN controllers handles several thousand messages per second (bridging / filtering / processing) with large headroom.
- The STM32↔ESP32 link runs at **115200 by default and has been verified at 1 Mbit/s** (8E1, interrupt-driven RX on the STM32). Streaming *every* CAN frame on to Wi-Fi gives roughly **580 frames/s at 115200** and **~5000 frames/s at 1 Mbit/s** (~18 bytes per framed CAN message).
- At 1 Mbit/s the link is therefore **no longer the bottleneck** — it can carry a fully saturated 500 kbit/s CAN bus (~4500 frames/s) live to Wi-Fi.

Firmware update & code protection (RDP1) — over Wi-Fi

The on-board ESP32-S3 is wired to the STM32's **NRST** and **BOOT0** lines, so it can put the MCU into its factory **ROM bootloader** (ST protocol AN3155) and reprogram it **entirely over Wi-Fi** — no ST-Link, no jumpers, no USB connection to the STM32.

STM32 signal	Driven by	Purpose
NRST	ESP32-S3 (also available on J4 pin 4)	hardware reset — enter the bootloader / restart the application
BOOT0	ESP32-S3 (10 kΩ pull-down on board)	selects the ROM bootloader at reset
USART3 (PB10 / PB11)	ESP32-S3	AN3155 bootloader protocol + normal data link

These three signals are routed **internally** between the two chips — nothing to wire up. This is what allows a complete remote firmware update of the STM32 from a browser.

✓ **RDP Level 1 read-out protection — enabled and removed over Wi-Fi.** After flashing, the STM32's **RDP1** read-out protection can be switched on directly from the web dashboard, so the firmware **cannot be read back** out of the chip. It can also be removed remotely when you need to reflash — all without a programmer.

- Protects your firmware / IP on boards you ship to customers.
- Repeatable: lock → later unlock → reflash → lock again, entirely over Wi-Fi.
- **Note:** removing RDP1 always performs a full **mass erase** of the STM32 flash — this is an STM32 hardware behaviour and is exactly what makes the protection effective.

J4 — SWD / ST-Link programming header

A 5-pin 2.54 mm header on the **right-hand edge** of the board. **Pin 1 is the pin closest to the top edge** (farthest from the power terminal J5); pin 5 is at the bottom.

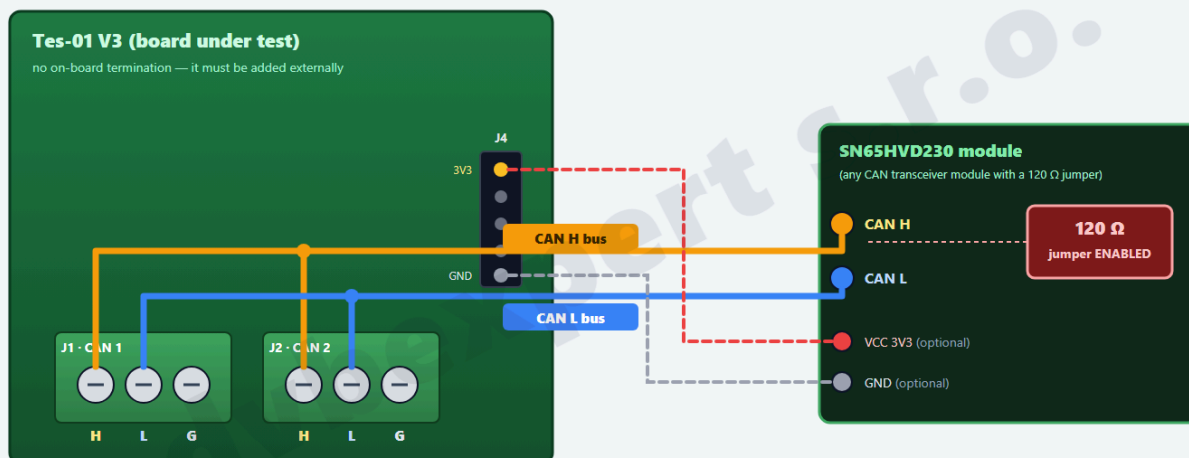
Pin	Signal	Connect to ST-Link (V2 / V3)
1 — top	3V3	3.3 V / VAPP (target-voltage sense)
2	SWDIO	SWDIO
3	SWCLK	SWCLK
4	NRST	RST / NRST (optional, recommended)
5 — bottom	GND	GND

Only needed for ST-Link / SWD programming or recovery. In normal use you never need it — the STM32 and the ESP32-S3 are both updated **over Wi-Fi** (and the ESP32 additionally over USB-C). Do not power the board from the ST-Link and from J5 at the same time.

Bench self-test setup (120 Ω termination)

The built-in CAN self-test can verify both channels on the bench. Bridge **J1 ↔ J2** (H ↔ H, L ↔ L) **and add a 120 Ω resistor between CAN H and CAN L** — any CAN transceiver module with its 120 Ω terminator enabled works too. Without this termination the self-test will fail with stuff/bit errors even though the board is fully functional.

Bench CAN self-test wiring — transceiver module as a 120 Ω terminator



How to wire it (2 wires + optional power)

1) J1 pin 1 (H) → J2 pin 1 (H) → module CAN H — all on one wire (amber branch)

2) J1 pin 2 (L) → J2 pin 2 (L) → module CAN L — second wire (blue branch)

3) on the module keep the **120 Ω jumper ENABLED** — this is the whole point: it terminates the bus

Optional: power the module from the board — J4 pin 1 (3V3) → module VCC. J4 pin 5 (GND) → module GND (dashed). The 120 Ω works unpowered too.

Warning: never connect CAN H and CAN L directly with a wire — only through the 120 Ω resistor.

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Firmware & software

Each board is delivered with **pre-loaded gateway firmware** providing:

- A **Wi-Fi web dashboard** (Access-Point or join your network) for live CAN monitoring and configuration
- **Over-the-air (OTA) firmware update** of both the ESP32-S3 and the STM32 — no programmer needed
- A built-in **CAN self-test** (loopback + cross-check) to validate both channels and wiring
- Optional read-out protection of the STM32 after programming

Custom programming service — on request. We can develop and load **custom firmware** for your specific CAN protocol, gateway logic, filtering, logging or telemetry requirements. Contact us with your specification for a quote.

What's included

- 1× Tes-01 board, fully assembled and tested, with the XIAO ESP32-S3 module fitted
- 1× 2.4 GHz Wi-Fi antenna (U.FL/IPEX)
- Pre-loaded gateway firmware + access to the web dashboard

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